UNIVERSITY OF CALIFORNIA SANTA CRUZ

The causal relationship between vowel devoicing and duration in Japanese and Korean

McGill UNIVERSITY

Hanyoung Byun (UC Santa Cruz) & **Seung Suk Lee** (McGill University)

hbyun5@ucsc.edu *seung.suk.lee@mcgill.ca*

1. Introduction

- Vowels are **shorter and more likely to devoice** in these contexts:
  (1) when they are **[+high]** (Lehiste 1970, Solé and Ohala 2010)
  (2) when they are before $C_{2[-voice]}$ (Chen 1970)

Is the short duration of devoiced vowels the *cause (H1)* or the *result (H2)* of devoicing?

H1: $V_{[+high]}$, $C_{2[-voice]}$ → Duration shortening → Vowel devoicing

- Vowel devoicing is a result of gestural overlap (Jun et al. 1997)
- Contexts affect vowel devoicing, mostly via vowel shortening

H2: $V_{[+high]}$, $C_{2[-voice]}$ → Vowel devoicing → Duration shortening

- In Tokyo Japanese, while both $C_{1[-voice]}V_{[\pm high]}$ shorten before $C_{2[-voice]}$, $C_{1[-voice]}\mathbf{V_{[+high]}}$ before $C_{2[-voice]}$ shortens much more (Tanner et al. 2019) because high vowels devoice in that context
- Contexts affect vowel shortening mostly via vowel devoicing

RQ1. Is duration of $C_{1[-voi]}V_{[+hi]}$ shorter than what is expected from the intrinsic duration of $V_{[+hi]}$ and shortening before $C_{2[-voi]}$?
RQ2. Is the likelihood of VD correlated with CV duration?
RQ3. Does devoicing mediate context effects on CV duration?

2. Methods

Tokens of interest: $C_{1[-voice]}\ V_{[\pm high]}\ C_{2[\pm voice]}$

Tokyo Japanese (TJ) (n= 116,876)

- **Corpus of Spontaneous Japanese** (Maekawa et al. 2000)
- $C_{2[+voice]}$ = {voiced obstruents, sonorants}
- $C_{2[-voice]}$ = {voiceless obstruents}
- V devoicing annotated in the corpus
- Long V and geminate C are excluded

Seoul Korean (SK) (n = 50,349)

- **Seoul Corpus** (Yun et al. 2015)
- $C_{2[+voice]}$ = {voiced lenis obstruents, sonorants}
- $C_{2[-voice]}$ = {voiceless fortis/aspirated obstruents}
- V devoicing, lenis voicing determined by Voice Report in Praat

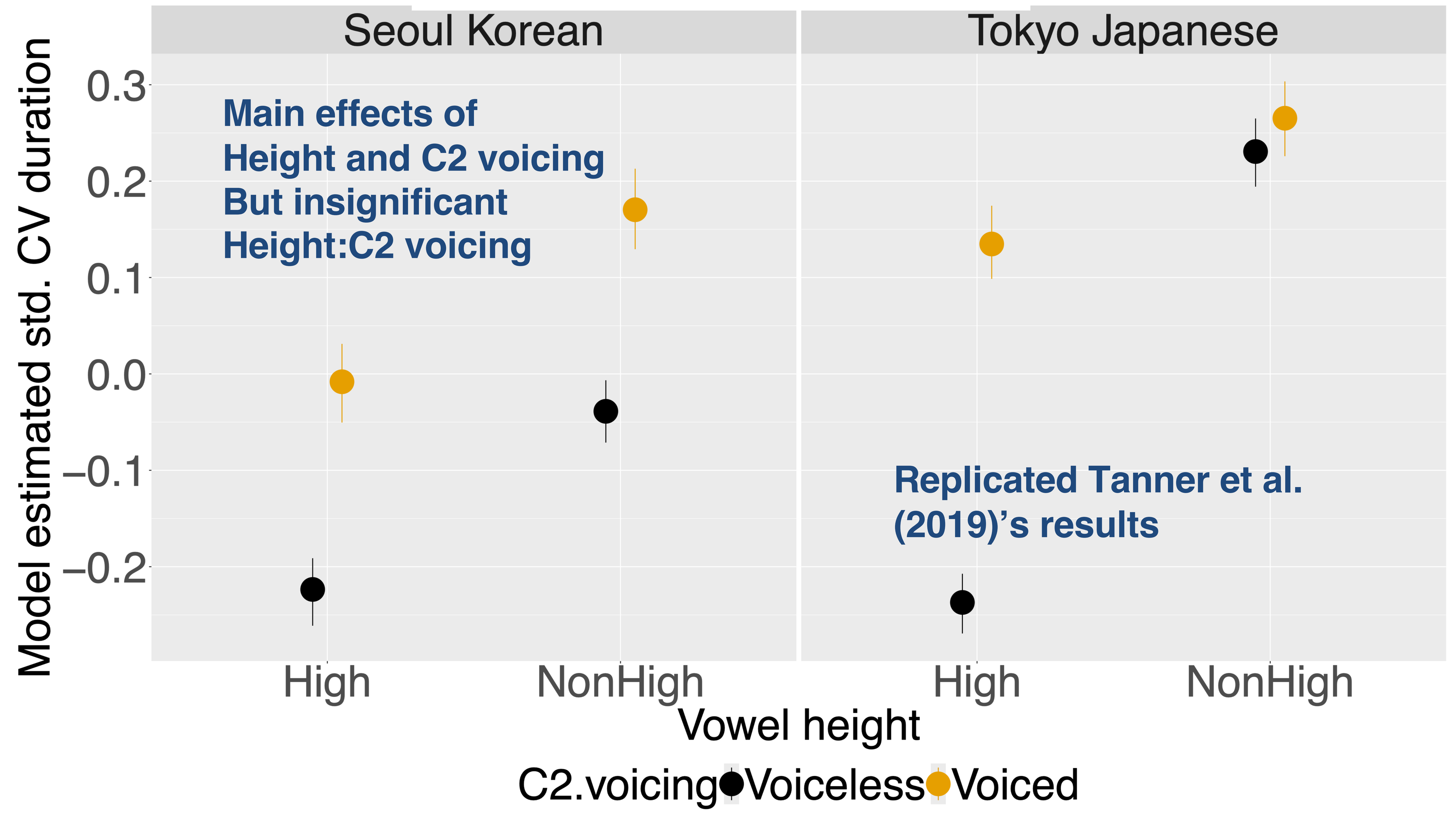
Baysian mixed-effects regression models

- DV: Vowel devoicing (Bernoulli with logit link), CV duration (Gaussian)
- IV: V height (binary), C2 voicing (binary)
- Covariates: Preceding C manner, Following C manner, speech rate
- by-speaker and by-word random intercepts and random slopes

Mediation analysis:
Compares regression models fit with and without a mediator variable (Cohen Priva and Gleason 2020, Katz and Pitzanti 2019, Lee 2025)

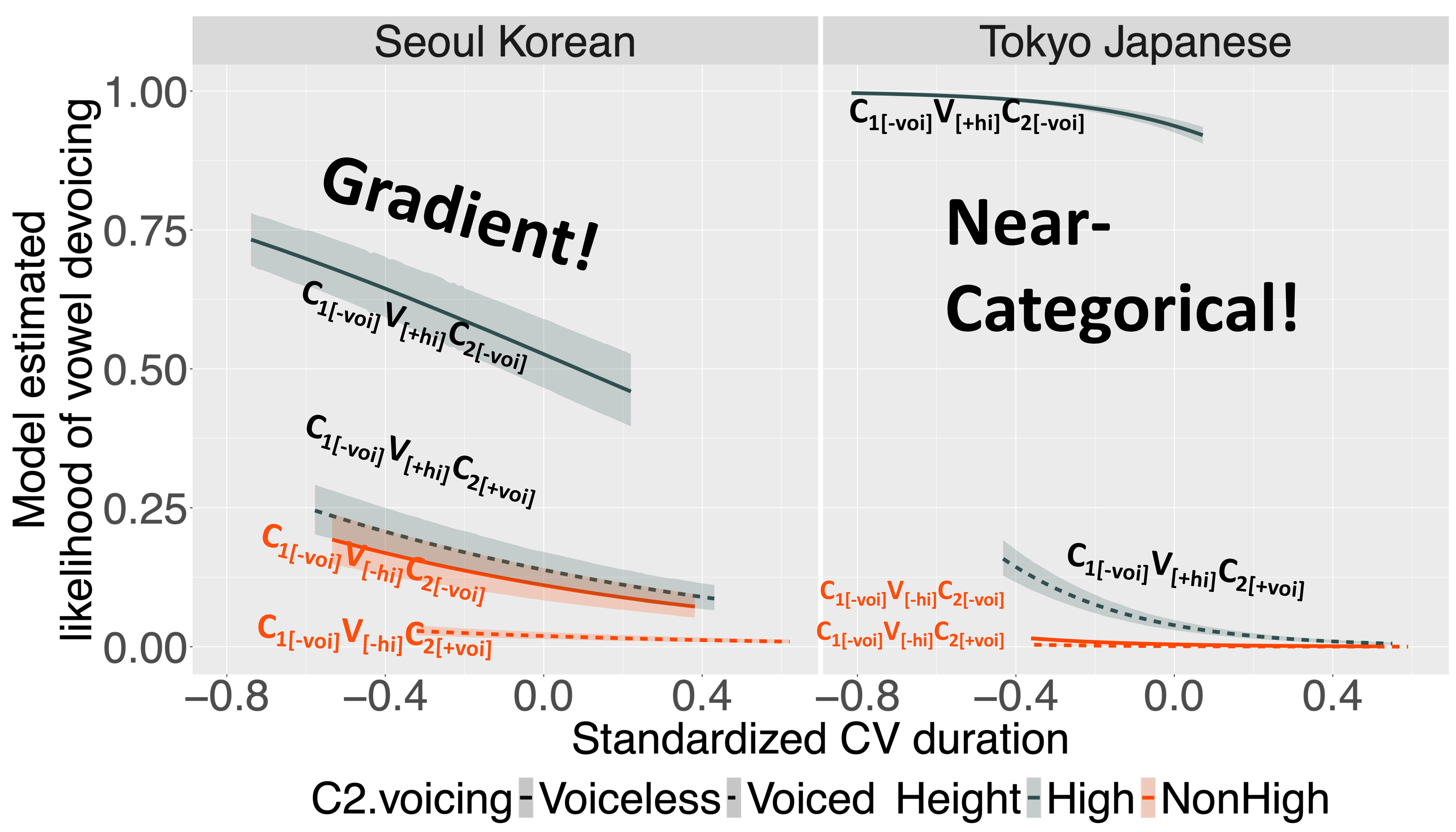
If context *causes* duration shortening mostly via vowel devoicing (H2), then the **indirect effect** on CV duration via vowel devoicing should be **similar to** the **total effect** of contexts on CV duration

3.1. CV duration



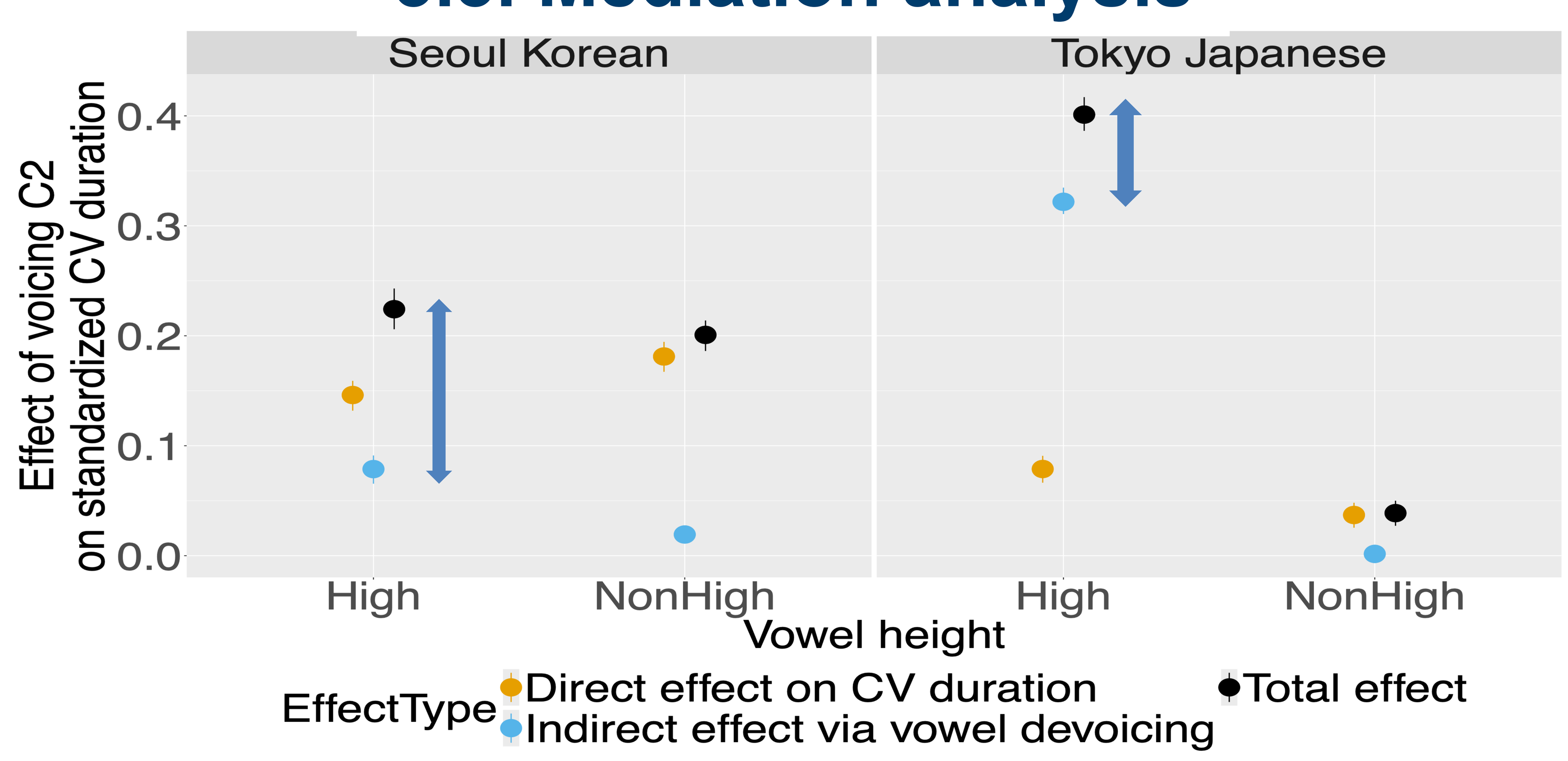
SK: Effect of shortening before $C_{2[-voice]}$ is **similar** across V heights
TJ: **$C_{[-voice]}V_{[+high]}$ becomes much shorter before $C_{2[-voice]}$** (most of which undergo VD) compared to $C_{[-voice]}V_{[-high]}$ before $C_{2[-voice]}$
↪**Supports H2 for TJ**

3.2. CV duration x Devoicing



SK: $C_{1[-voice]}V_{[+high]}C_{2[+voice]} \approx C_{1[-voice]}V_{[-high]}C_{2[-voice]}$
↪**Supports H1 for SK**
$C_{1[-voice]}V_{[+high]}C_{2[-voice]} > C_{1[-voice]}V_{[-high]}C_{2[-voice]}$
↪**Challenges H1 for SK**
TJ: Very weak correlation between CV duration and the likelihood of VD
↪**Challenges H1 for TJ**

3.3. Mediation analysis



SK: The indirect effect via vowel devoicing is ***smaller*** than the direct effect of C2 voicing on CV duration
↪ **Challenges H2 for SK**
TJ: The indirect effect of C2 voicing on CV duration via vowel devoicing is ***similar*** to the **total effect of C2 voicing on CV duration**
↪ **Supports H2 for TJ**
↪ Devoicing ***mediates*** context effects on CV duration

4. Discussion

Our results
TJ: Strong support for H2 (Context → VD → Duration shortening)
SK: Weak support for H1 (Context → Duration shortening → VD)

- Potential reason for the difference between SK and TJ: **Phonologization**
- Coarticulation as a precursor to a phonological process (Beddor 2009)
- At the **phonetic** level, VD occurs as a result of coarticulatory shortening (and other factors), as predicted by gestural overlap (H1)
- But once VD is **phonologized**, short duration of CV cannot be attributed to coarticulatory shortening (H2)
- Phonologization may change the causal structure of a process and its associated phonetic variables (H1 → H2)

5. Conclusion

- **Variation in CV duration** can be explained by general coarticulatory shortening in SK, but **not in TJ.**
- **The likelihood of VD** can be explained by coarticulatory shortening in SK to some extent, but **not in TJ**.
- The difference could be due to **phonologization** —VD is phonologized in TJ while VD is at the phonetic level in SK.

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Acknowledgements: Many thanks to Ryan Bennett, Morgan Sonderegger, Kristine Yu, and members of P* reading group at McGill University for their helpful feedback. All errors are our own.